

CG LAW RAG:
A Retrieval-Augmented Generation System
for U.S. Coast Guard Legal Information

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1. Project Overview and Motivation

Coast Guard law and policy are distributed across several documents spanning thousands of pages. These documents cover UCMJ criminal offenses, Coast Guard conduct, administrative separation policy, and substance abuse programs. Junior officers and cadets must manually search these materials to answer legal questions, which is inefficient and prone to human error. Coast Guard junior officers and cadets are also required to write Preliminary Inquiry Officer (PIO) reports when legal incidents occur involving service members. These reports are time-consuming, requiring a summary of events, identification of applicable law and policy, and recommendations for disposition.

Our original proposal was to develop a Large Language Model (LLM) capable of drafting PIO reports from raw investigative inputs (witness statements and appointing memoranda). We identified candidate base models, including AdaptLLM/law-LLM and SaulLM-7B, and designed a fine-tuning pipeline. However, this project was infeasible due to a lack of training data. We had access to only about 25 PIOs. This is far fewer than the thousands of labeled examples required for reliable fine-tuning in a specialized legal domain. We could not expand our dataset because PIOs contain personally identifiable information (PII), and obtaining approval to use them is a lengthy process, which was not feasible within a single semester. As a result, we pivoted to a retrieval-augmented generation (RAG) system. This system enables faster and more accurate legal information retrieval, significantly reducing the time required to complete PIO reports while avoiding the need for sensitive training data.

This project was developed to address that problem: a system that receives a legal question, retrieves the relevant sections from the appropriate documents, and summarizes the answer with full source citations—without acting as a substitute for a licensed attorney.

Business Problem Statement: Coast Guard personnel spend unnecessary time manually searching multiple legal manuals for answers that should be accessible within seconds. The risk of misinterpreting policy due to incomplete searches affects cadets, junior officers, and operational

units. This project evaluates whether citation-grounded AI can improve access to legal information without replacing legal professionals or providing independent legal advice.

Analytics Problem Statement: Build a RAG pipeline that ingests U.S. Coast Guard directives and federal regulations, indexes them for semantic retrieval, and generates plain-language summaries grounded strictly in the source text with citations to the exact policy passage. Performance is evaluated using Recall@ k , Precision@ k , F1@ k , and Mean Reciprocal Rank (MRR).

2. Evolution of the Project

Phase 1: MCM-Only Prototype

We began with a narrow scope. Rather than ingesting every relevant document at once, we selected the Manual for Courts-Martial (MCM) 2024 Punitive Articles (Part IV, Articles 77–134) as the initial corpus. The pipeline consisted of four scripts: document extraction and chunking, FAISS vector index construction, query-based retrieval, and benchmark evaluation.

The system indexed 370 chunks across 58 UCMJ articles using all-MiniLM-L6-v2 (384-dimensional embeddings) and supported dense nearest-neighbor retrieval via FAISS IndexFlatIP. A 10-question benchmark confirmed that basic semantic queries worked. However, queries involving exact legal terminology, such as “hazing,” performed poorly because the general-domain embedding model does not capture domain-specific mappings (e.g., “hazing” corresponds to Article 93 in the MCM).

Phase 2: Corpus Expansion and Hybrid Retrieval

In Phase 2, we expanded the dataset and improved the retrieval process. We added the Conduct Manual, Investigations Manual, Correspondence Manual, Separations Manual, and Substance Abuse Manual. This increased the corpus from 370 to 1,171 chunks.

We also introduced BM25 sparse retrieval and combined it with dense retrieval using Reciprocal Rank Fusion (RRF). This change produced the most significant improvement in system perfor-

mance. Without modifying the LLM, adding BM25 resulted in an 8-percentage-point increase in Recall@1 and a 0.083 improvement in Mean Reciprocal Rank (MRR). This gave us an evident improvement in performance. We concluded that the quality of the retrieval process is crucial for an effective RAG system.

Finally, further improvements were made to our retrieval process by increasing the benchmark dataset from 10 to 200 questions (with seven categories spanning across all document content).

3. Data and Preprocessing

The Corpus

Six official documents were ingested, totaling 1,171 chunks:

Document	Citation	Chunks
MCM 2024 Punitive Articles	Part IV, Arts. 77–134	~292
CG Conduct Manual	COMDTINST M1600.2	~158
CG Admin. Investigations Manual	COMDTINST M5830.1A	~189
CG Military Separations Manual	COMDTINST 1000.4C	~218
CG Substance Abuse Manual	COMDTINST 1000.10B	~52
CG Military Justice Manual	COMDTINST M5810.1H	~262
Total		1,171

Chunking and Preprocessing Challenges

Each document was split at natural section or article boundaries using regular expressions, then subdivided into approximately 600-token chunks with 80–100 token overlap to preserve context. Each chunk stores metadata including source label, section number, heading hierarchy, PDF page range, content hash, and a unique identifier.

Three major preprocessing challenges arose:

1. **PDF Artifacts:** Page markers and hyphenation were removed using regex cleaning.
2. **Ambiguous Section Labels:** Repeated section labels (e.g., A., B., C.) across documents caused evaluation mismatches, resolved by adding a `source.doc` constraint.
3. **Tokenization Differences:** BM25 treats hyphenated terms differently than natural queries (e.g., “line-of-duty”), requiring dense retrieval to compensate for semantic matching.

4. Modeling Approach

Dense Retrieval Baseline

Queries and chunks are embedded using all-MiniLM-L6-v2 and L2-normalized. An embedding is a vector of numbers that captures the semantic meaning of a piece of text. All-miniLM-L6-v2 is a model that produces 384-dimensional embeddings and was trained on over a billion sentence pairs to capture a rich representation of language. We evaluated alternative embedding models, including BAAI/bge-small-en-v1.5 and bge-base-en-v1.5, but retained MiniLM for consistency with benchmark comparisons. Retrieval uses cosine similarity via FAISS:

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \mathbf{q} \cdot \mathbf{d} \quad (\text{cosine, since both vectors are L2-normalized})$$

This compares vectors by their direction, so the system identifies document chunks whose meaning is most similar to the user’s query and retrieves those most relevant results.

BM25 Sparse Retrieval

BM25 assigns a relevance score to each document based on term frequency, inverse document frequency, and length normalization, which is used to determine the rank of the documents and retrieve the most relevant matches. We implemented BM25Okapi with a custom tokenizer preserving legal identifiers such as “112a.”

A 12-rule query expansion system maps common terms (e.g., “hazing”) to formal legal terminology (e.g., “cruelty maltreatment Article 93”).

$$\text{BM25}(D, Q) = \sum_i \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}} \right)}$$

where $k_1 = 1.5$, $b = 0.75$, and IDF is:

$$\text{IDF}(q_i) = \log \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right)$$

Hybrid Retrieval via Reciprocal Rank Fusion

Dense and BM25 ranked lists are fused using Reciprocal Rank Fusion (RRF), which uses only rank rather than raw scores, avoiding any cross-system score normalization:

$$\text{RRF}(c) = \frac{1}{k + r_{\text{dense}}(c)} + \frac{1}{k + r_{\text{BM25}}(c)}, \quad k = 60$$

A chunk ranked first in both systems scores about 0.033; ranked first in only one, about 0.016. This approach rescued queries where exact sub-article numbers like 87a and 112a returned zero dense results, because BM25 exact matching filled the gap. This method avoids score normalization and improves robustness, particularly for exact legal queries. RRF improved the retrieval process of our system because it ensured relevant documents were ranked by a combination of semantic similarity and keywords.

LLM Generation

Mistral 7B Instruct is used as the summarization layer. Retrieved passages are structured into:

- Plain-language summary
- Key elements

- Maximum punishment
- Citations
- Legal disclaimer

Mistral 7B is given the top- k relevant documents as the context and uses this to generate an answer to the query. The model is constrained to use only retrieved information.

The role of the LLM in this system is to transform the retrieved legal information into a clear and useful summary. The retrieval process returns the relevant chunks, and the LLM uses this context to provide a clear, plain-language answer to the user’s query. This allows the model to provide accessible, user-friendly outputs while remaining fully grounded in the original legal sources.

5. Evaluation and Results

Metrics

We evaluated at $k \in \{1, 3, 5, 8\}$, where k is the number of top results returned by the retrieval system, using four retrieval metrics (each defined below) across a 200-question benchmark.

Recall@ k measures whether the correct answer appeared anywhere in the top- k results. It is the primary metric because missing the right answer entirely is worse than including some irrelevant ones:

$$\text{Recall@}k = \frac{|\text{Relevant} \cap \text{Retrieved}_k|}{|\text{Relevant}|}$$

Precision@ k measures what fraction of the top- k retrieved results were actually relevant:

$$\text{Precision@}k = \frac{|\text{Relevant} \cap \text{Retrieved}_k|}{k}$$

F1@ k is the harmonic mean of Precision@ k and Recall@ k , giving a single combined score:

$$\text{F1@}k = \frac{2 \cdot \text{Precision@}k \cdot \text{Recall@}k}{\text{Precision@}k + \text{Recall@}k}$$

Mean Reciprocal Rank (MRR) rewards systems that rank the correct answer near the top. It is the average of $1/\text{rank}$ of the first relevant result across all queries:

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$

The 200-question benchmark covered policy lookups (50 questions), definitions (41 questions), procedures (35 questions), article lookups (33 questions), elements of offenses (25 questions), cross-document questions (9 questions), and maximum punishment questions (7 questions).

Dense vs. Hybrid Results

Metric	Dense	Hybrid	Change
Recall@1	52.5%	60.5%	+8.0 pp
Recall@3	67.5%	79.0%	+11.5 pp
Recall@5	77.0%	84.5%	+7.5 pp
Recall@8	84.5%	90.0%	+5.5 pp
Precision@1	52.5%	60.5%	+8.0 pp
Precision@3	40.2%	47.0%	+6.8 pp
Precision@5	32.5%	37.7%	+5.2 pp
Precision@8	26.4%	30.3%	+3.9 pp
F1@1	52.5%	60.5%	+8.0 pp
F1@5	45.7%	52.1%	+6.4 pp
MRR	0.625	0.708	+0.083

As shown in the results table above, hybrid retrieval improved every metric. The Recall@1 im-

provement is especially significant, as users rely most on top-ranked results. Remaining failures involved complex cross-document queries and low-confidence retrieval cases.

6. Final Model and Interpretation

The final system is a hybrid BM25 + dense RAG pipeline with LLM summarization. Queries are embedded, relevant documents are retrieved via FAISS and BM25, fused using RRF, and the LLM generates an answer with citations.

RRF scores are interpretable: values near 0.033 indicate top ranking in both systems, while ~ 0.016 indicates top ranking in one.

Using $k = 5$ ($\text{Recall}@5 = 84.5\%$), the system reliably surfaces relevant sources for most queries. It performs best on focused legal lookups and less reliably on cross-document questions.

This model is the first step to automated PIO generation and serves as a significant milestone in the progression to that final goal. The retrieval process was an important first step; the next step is to constrain the output to the PIO report format when we have enough data to train a model.

7. Reflection and Future Work

What We Would Do Differently

An assessment of the amount of data we had compared to the amount of data required would have prevented time spent on an infeasible PIO-generation LLM. Additionally, building the evaluation benchmark earlier would have improved development efficiency. We should have focused on improving just the retrieval accuracy of the system first before moving on to the generation aspect.

Next Steps

- Upgrade to BAAI/bge-small embeddings
- Evaluate final RAG answer (using LLM-as-a-judge or manually evaluate final model output)

- Develop a FastAPI backend and web interface

What We Learned

Retrieval quality is the primary bottleneck in RAG systems. Hybrid retrieval significantly outperformed dense-only methods. The evaluation metrics were essential for identifying system weaknesses and finding areas of improvement.

PDF extraction from two-column legal documents was the most challenging technical issue we faced. The library `pypdf` that we used does not extract cleanly from print-formatted documents, so residual artifacts get mixed into chunk text alongside legal content. This corrupts the vector embeddings for that chunk, which can decrease a result's ranking.

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